KAIZEN MATHS

Lesson starter

Complex Numbers

Loci and Exam-Style Problems

Name _____

S1

Describe each locus on the Argand diagram.

Describe the locus $|z - (-2)| = 6$ on the Argand diagram.

S2

Find the Cartesian equation of each locus.

Let $z = x + iy$. Find the Cartesian equation of the locus $|z - (-1 + 4i)| = |z - (1 + 4i)|$.

S3

Describe each argument locus.

Describe the locus $\arg(z - (3 + 4i)) = -\pi/4$.

S4

Describe each locus on the Argand diagram.

Describe the locus $|z - (5)| = 6$ on the Argand diagram.

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ALG-D7A3-L4 · S

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KAIZEN MATHS

Exit ticket

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Name

E1

Find the Cartesian equation of each locus.

Let z = x + iy. Find the Cartesian equation of the locus $|z - (-i)| = |z - (8i)|$.

E2

Describe each argument locus.

Describe the locus $\arg(z - (4 - 2i)) = \pi/6$.

E3

Describe each locus on the Argand diagram.

Describe the locus $|z - (-4 + 5i)| = 4$ on the Argand diagram.

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Show your working

Teacher answers & working

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S1

Describe each locus on the Argand diagram.

Describe the locus $|z - (-2)| = 6$ on the Argand diagram.

Answer:

Circle with centre (-2, 0) and radius 6

Working:

Let $z = x + iy$.

The expression $|z - (-2)|$ measures the distance from z to -2.

The fixed point -2 has coordinates (-2, 0) on the Argand diagram.

All points a distance 6 from (-2, 0) form a circle.

Therefore the locus is a circle with centre (-2, 0) and radius 6.

S2

Find the Cartesian equation of each locus.

Let $z = x + iy$. Find the Cartesian equation of the locus $|z - (-1 + 4i)| = |z - (1 + 4i)|$.

Answer:

$x = 0$

Working:

The point z is the same distance from -1 + 4i and 1 + 4i.

So the locus is the perpendicular bisector of the line segment joining the two fixed points.

The fixed points are (-1, 4) and (1, 4).

They have the same y-coordinate, so the perpendicular bisector is vertical.

$x = \frac{-1+1}{2} = 0$.

S3

Describe each argument locus.

Describe the locus $\arg(z - (3 + 4i)) = -\pi/4$.

Answer:

A ray from (3, 4) making angle $-\pi/4$ with the positive real direction

Working:

$z - (3 + 4i)$ translates the origin to the point (3, 4).

$\arg(z - (3 + 4i))$ is the angle made from that point.

The argument is fixed at $-\pi/4$.

Therefore the locus is a ray starting at (3, 4) and making angle $-\pi/4$ with the positive real direction.

The endpoint (3, 4) is not included because the argument is undefined there.

Teacher answers & working

Complex Numbers · Loci and Exam-Style Problems

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S4

Describe each locus on the Argand diagram.

Describe the locus |z - (5)| = 6 on the Argand diagram.

Answer:

Circle with centre (5, 0) and radius 6

Working:

Let $z = x + iy$.

The expression |z - (5)| measures the distance from z to 5.

The fixed point 5 has coordinates (5, 0) on the Argand diagram.

All points a distance 6 from (5, 0) form a circle.

Therefore the locus is a circle with centre (5, 0) and radius 6.

E1

Find the Cartesian equation of each locus.

Let z = x + iy. Find the Cartesian equation of the locus |z - (-i)| = |z - (8i)|.

Answer:

$y = 3.5$

Working:

The point z is the same distance from -i and 8i.

So the locus is the perpendicular bisector of the line segment joining the two fixed points.

The fixed points are (0, -1) and (0, 8).

They have the same x-coordinate, so the perpendicular bisector is horizontal.

$y = \frac{-1+8}{2} = 3.5$.

E2

Describe each argument locus.

Describe the locus arg(z - (4 - 2i)) = π/6.

Answer:

A ray from (4, -2) making angle π/6 with the positive real direction

Working:

z - (4 - 2i) translates the origin to the point (4, -2).

arg(z - (4 - 2i)) is the angle made from that point.

The argument is fixed at π/6.

Therefore the locus is a ray starting at (4, -2) and making angle π/6 with the positive real direction.

The endpoint (4, -2) is not included because the argument is undefined there.

Teacher answers & working

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E3

Describe each locus on the Argand diagram.

Describe the locus $|z - (-4 + 5i)| = 4$ on the Argand diagram.

Answer:

Circle with centre (-4, 5) and radius 4

Working:

Let $z = x + iy$.

The expression $|z - (-4 + 5i)|$ measures the distance from z to -4 + 5i.

The fixed point -4 + 5i has coordinates (-4, 5) on the Argand diagram.

All points a distance 4 from (-4, 5) form a circle.

Therefore the locus is a circle with centre (-4, 5) and radius 4.